SOFTWARE TESTING REPORT

Hospital Management System

UNIT	INTEGRATION	SYSTEM

Prepared for Software Engineering Mini-Project submission

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Execution date: 11 November 2026 | Asia/Calcutta

Assessment status: Executed automated scope passed; full matrix partially executed

Document Control and Executive Result

Item	Value
Document	Software Testing Report
Project	Hospital Management System
Version	1.0 - Final execution evidence
Execution date	11 November 2026
Prepared from	Local repository, requirements documents, source code, CI configuration, automated logs, database records, and browser evidence

64	0	20	76.2%
Passed	**Failed**	**Not executed**	Full-matrix pass

All 64 executed cases passed after four application defects were corrected and regression-tested. Twenty requested UI/manual cases remain NOT EXECUTED; therefore, this report does not declare the entire project fully passed. It concludes that the executed automated scope passed and the remaining scope requires manual or extended environment testing.

1. Project Introduction

The Hospital Management System is a role-based web application built with React and Vite on the client, Express and Mongoose on the server, and MongoDB for persistence. It supports Admin, Receptionist, Doctor, Patient, and Guest workflows across identity, departments, clinical operations, facilities, admissions, billing, laboratory records, notifications, and reporting.

The assessment began with the repository README, SRS, existing test plan, coverage notes, final verification checklist, package manifests, CI workflow, routes, models, frontend pages, and existing tests. Features were treated as unverified until exercised by a test or explicitly marked not executed.

2. Objectives of Testing

- Validate core domain rules at unit level, including boundary and invalid-input behavior.
- Verify API, authentication, role authorization, database persistence, scheduling, facilities, billing, clinical records, laboratories, notifications, and reporting against an isolated MongoDB database.
- Exercise representative end-to-end journeys through the real React interface using Chrome automation.
- Detect and correct genuine defects without weakening assertions or changing unrelated functionality.
- Produce reproducible evidence and distinguish PASS, FAIL, BLOCKED, and NOT EXECUTED outcomes.

3. Scope of Testing

In scope	Assessment approach
Unit	Validation helpers, models, scheduling/availability helpers, billing arithmetic, notification rules, middleware authorization, and source wiring.
Integration	HTTP/API behavior with MongoDB, authentication, RBAC, CRUD, appointment conflicts, admissions/beds, discharge, clinical history, laboratory records, billing/payment, notifications, reports, and persistence reads.
System	Role login and dashboards, department/doctor/patient UI actions, reports, role denial, doctor/patient portal visibility, guest booking, duplicate booking prevention, and mobile layout.
Regression	Unit, integration, frontend build, backend syntax, and complete 12-case browser suite after fixes.

4. Software and Hardware Test Environment

Component	Configuration
Operating system	Microsoft Windows NT 10.0.26300.0, 64-bit process
Runtime	Node.js v22.20.0; npm 10.9.3
Database	MongoDB 8.2.1 Windows service
Browser automation	Playwright 1.62.1 with Chrome 157.0.8103.50
Frontend	React 18.3.1; Vite 6.4.5
Backend	Express 4.21.2; Mongoose 8.9.5
Integration database	hms_testing_report_20261111
System database	hms_system_testing_20261111
Hardware	Intel64 Family 6 Model 186; RAM not exposed by managed environment
Unavailable	Docker and Computer Use browser host; neither prevented execution because local MongoDB and bundled Playwright were available

Database safety: all integration and system operations used dedicated databases. No normal development or production database was targeted.

5. Testing Methodology

- Black-box API and UI assertions were combined with white-box unit tests for validation and domain helpers.
- Positive, negative, boundary, duplicate/conflict, authentication, and authorization cases were included.
- Integration cases verified status codes and response data and performed read-back checks from MongoDB-backed endpoints.
- System tests drove actual forms and routes, recorded screenshots, and failed on uncaught browser page errors.
- Defects were classified, corrected, and followed by targeted and full regression runs.

6. Unit Testing

Metric	Result
Command	cd backend; npm test
Discovered	47 Node test entries
Passed	44
Failed	0
Skipped	3 opt-in integration bootstrap files; excluded from unit total
Duration	977.9252 ms
Evidence	Test_Execution_Logs/unit_tests.log

Coverage classification

Area	Verified behavior
Identity validation	Email, username, password length, valid and invalid authentication-shaped inputs
Patient and doctor	Required fields, email/phone validation, doctor duration and availability
Departments and appointments	Department constraints, slot boundaries, duration alignment, overlap and working hours
Admissions and facilities	Admission validity, active-bed uniqueness, discharge rules, bed allocation behavior
Billing and payment	Subtotal/discount/tax/total arithmetic, payment validity, remaining balance
Clinical and notification	Clinical record requirements, notification generation and reminder rules
Authorization	Role middleware allow/deny behavior and source-wiring checks

The final unit suite adds focused domain coverage without duplicating existing cases. Valid, invalid, and boundary examples are included. The raw Node output remains the authority for individual assertions.

7. Integration Testing

Metric	Result
Command	npm run test:integration with all three opt-in flags enabled
Database	hms_testing_report_20261111
Passed	8
Failed	0
Skipped	0
Duration	11695.0931 ms
Evidence	Test_Execution_Logs/integration_tests.log

ID	Scenario	Status
IT-01	Authentication, accounts, departments, doctors and patients	PASS
IT-02	Appointments, availability, duplicate-slot protection and persistence	PASS
IT-03	Admissions, occupied-bed rejection, discharge and bed release	PASS
IT-04	Clinical history and patient-record access control	PASS
IT-05	Billing, payments, totals, patient scope and persistence	PASS
IT-06	Laboratory records and patient linkage	PASS
IT-07	Notification generation and appointment reminder behavior	PASS
IT-08	Reports and database aggregation	PASS

The Node test runner reports eight top-level integration tests. Several top-level cases contain multiple endpoint and persistence assertions. Unauthorized access and duplicate/conflicting resource operations are explicitly asserted.

8. System Testing

The production-like browser assessment used the actual React UI at localhost:5174 and the Express API at localhost:5200, backed by the isolated hms_system_testing_20261111 database. The final regression completed on 11 November 2026 at 13:24 local time.

Test ID	Description	Actual result	Status
ST-AUTH-NEG-01	Invalid staff login displays an error	Invalid credentials were rejected with the expected visible error.	PASS
ST-ADMIN-01	Admin login and dashboard	Admin dashboard loaded after credential authentication.	PASS
ST-ADMIN-DEPT-01	Admin creates, updates, and deactivates a department	Department create, rename, and deactivate operations persisted through the UI.	PASS
ST-ADMIN-DOCTOR-01	Admin creates, updates availability, and deactivates a doctor	Doctor account creation, 45-minute duration update, and deactivation succeeded.	PASS
ST-ADMIN-PATIENT-01	Admin registers and updates a patient	Patient registration produced a Patient ID and the updated phone persisted.	PASS
ST-ADMIN-REPORTS-01	Admin reports page loads current persisted data	Reports workspace rendered for Admin with live summary content.	PASS
ST-RECEPTION-01	Receptionist login and Admin route denial	Receptionist dashboard loaded and direct Admin report access redirected to the role dashboard.	PASS
ST-DOCTOR-01	Doctor login, dashboard, appointments, and patient workspace	Doctor-specific dashboard, own appointments, and assigned-patient pages rendered.	PASS
ST-PATIENT-01	Patient login and scoped portal pages	Patient dashboard, medical records, prescriptions, and billing pages rendered for the linked profile.	PASS
ST-GUEST-01	Guest books an appointment without creating a login	Guest confirmation rendered; patient record persisted; login remained unavailable.	PASS
ST-GUEST-NEG-01	Duplicate guest slot is rejected	Second booking received HTTP 409: That time slot overlaps an existing appointment. Please choose another time.	PASS
ST-RESPONSIVE-01	Guest booking page renders at mobile viewport	Mobile viewport rendered without horizontal document overflow.	PASS

Final browser result: 12 passed, 0 failed. The run produced 11 genuine screenshots. It recorded no uncaught page errors and no invalid HTML-pattern errors. Expected console noise from intentionally rejected 401/404 requests and sandbox-blocked external resources remains visible in the JSON evidence.

9. Functional Test Cases and Unexecuted Scope

The workbook is the complete row-level test register. The following table records the requested workflows that were not exercised end-to-end through the browser in this assessment. They are not counted as passes.

ID	Role/area	Scenario	Status
ST-NE-01	Admin	Ward management UI CRUD	NOT EXECUTED
ST-NE-02	Admin	Room management UI CRUD	NOT EXECUTED
ST-NE-03	Admin	Bed management UI CRUD	NOT EXECUTED
ST-NE-04	Admin	Admission management through UI	NOT EXECUTED
ST-NE-05	Admin	Discharge management through UI	NOT EXECUTED
ST-NE-06	Admin	Billing and payment mutation through UI	NOT EXECUTED
ST-NE-07	Reception	Patient delete with active-record restrictions	NOT EXECUTED
ST-NE-08	Reception	Book appointment through receptionist UI	NOT EXECUTED
ST-NE-09	Reception	Reschedule appointment through receptionist UI	NOT EXECUTED
ST-NE-10	Reception	Cancel appointment through receptionist UI	NOT EXECUTED
ST-NE-11	Reception	Register admission through receptionist UI	NOT EXECUTED
ST-NE-12	Doctor	Update availability and duration through doctor UI	NOT EXECUTED
ST-NE-13	Doctor	Complete consultation and add diagnosis/treatment	NOT EXECUTED
ST-NE-14	Doctor	Add prescriptions through UI	NOT EXECUTED
ST-NE-15	Doctor	Update doctor profile through UI	NOT EXECUTED
ST-NE-16	Patient	Book/reschedule/cancel appointment through patient UI	NOT EXECUTED
ST-NE-17	Patient	Update patient profile through UI	NOT EXECUTED
ST-NE-18	Output	Browser print/receipt dialog verification	NOT EXECUTED
ST-NE-19	Concurrency	Simultaneous independent human UI sessions	NOT EXECUTED
ST-NE-20	Performance	Large 12-month production-volume benchmark	NOT EXECUTED

10. Non-functional and Regression Results

Check	Method	Actual result	Status
Input validation	Unit/API/UI	Valid, invalid and boundary values exercised; final HTML patterns valid	PASS
Authentication security	API/UI	Invalid credentials rejected; JWT-protected routes enforced	PASS
Role authorization	API/UI	Unauthorized patient billing scope fixed; receptionist Admin route denied	PASS
Data integrity	MongoDB	Duplicate appointment and occupied-bed conflicts rejected; persistence read back	PASS
API response handling	Integration	Status codes and response bodies asserted across modules	PASS
Responsive layout	Chrome 390px	Guest page rendered without horizontal document overflow	PASS
Frontend production build	Vite	61 modules transformed; build completed in 2.25 s	PASS
Backend syntax	node --check	39 JavaScript source files passed	PASS
Load/security certification	Specialized tooling	Not performed; no certification claim	NOT EXECUTED

Regression sequence

- Reran unit suite: 44 unit passes, no failures.
- Reran isolated integration suite: 8 passes, no failures.
- Reran frontend production build after the final pattern correction: successful.
- Reran all 12 browser cases: all passed, with no uncaught page errors or pattern diagnostics.

11. Test Execution Summary

Testing type	Total	Passed	Failed	Blocked	Not executed	Pass %
Unit Testing	44	44	0	0	0	100.0%
Integration Testing	8	8	0	0	0	100.0%
System Testing	32	12	0	0	20	37.5%
Overall	84	64	0	0	20	76.2%

Pass percentage uses Passed / Total. Blocked and not-executed cases are not counted as passed. The executed-only pass rate is 64/64 = 100%.

12. Defect Report

ID	Severity	Defect	Classification	Resolution / verification	State
DEF-001	High	Patient could override billing scope with another patient ID	Application authorization bug	Route enforces linked patient identity and returns 403; integration regression added	FIXED
DEF-002	High	Doctor and Patient dashboards crashed: PageHeader not defined	Frontend import/export bug	Exported shared PageHeader and imported it in both dashboards	FIXED
DEF-003	Medium	Username/phone pattern attributes invalid under browser v-mode	Frontend validation bug	Escaped Unicode-set special characters; final system run clean	FIXED
DEF-004	Low	Duplicate Admission bed index declaration produced Mongoose warning	Schema definition issue	Removed redundant index flag while preserving partial unique active-bed index	FIXED

13. Screenshots and Execution Evidence

The screenshots below were generated by the final Playwright system suite. Original full-resolution PNG files are retained in Documents/TestingReport/Screenshots. Terminal and JSON evidence is retained in Test_Execution_Logs.

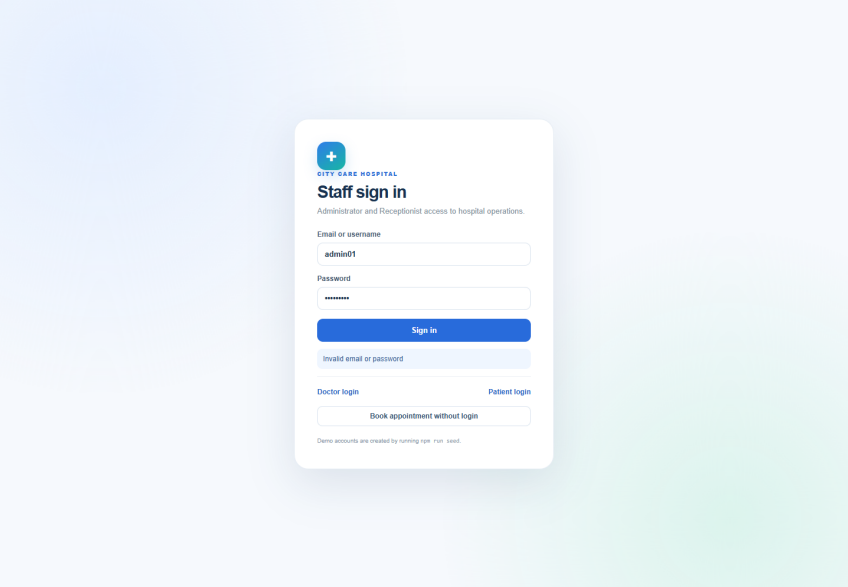


Figure 1. Invalid credentials rejected with a visible error.

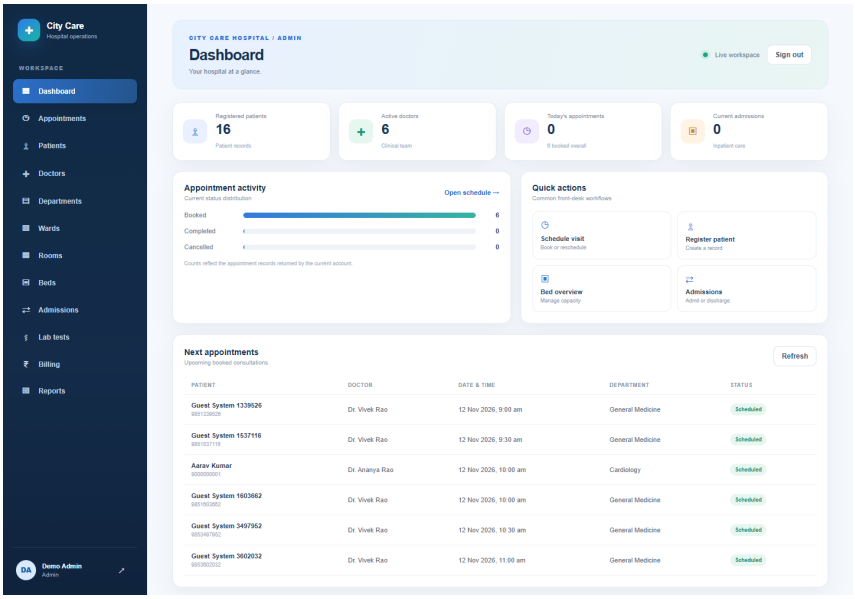


Figure 2. Admin dashboard after successful authentication.

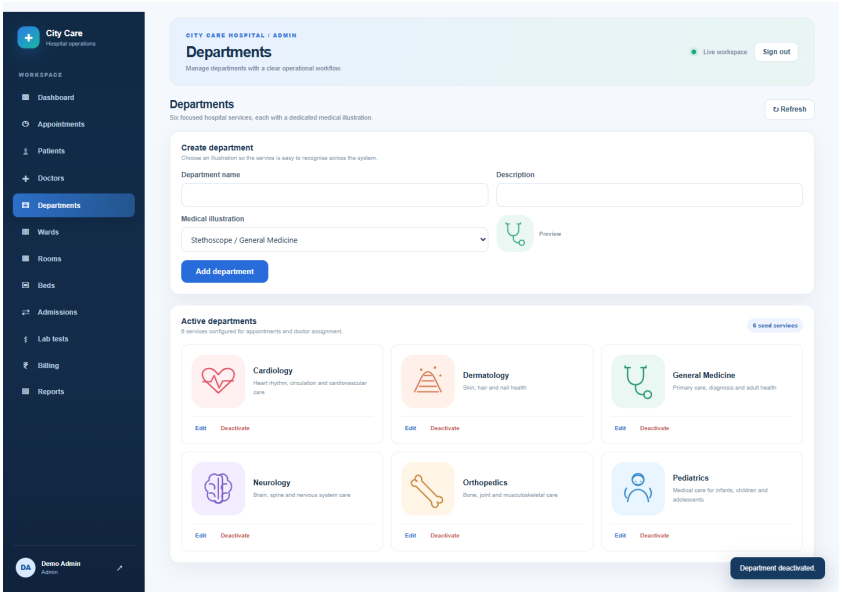


Figure 3. Department create/update/deactivate browser workflow.

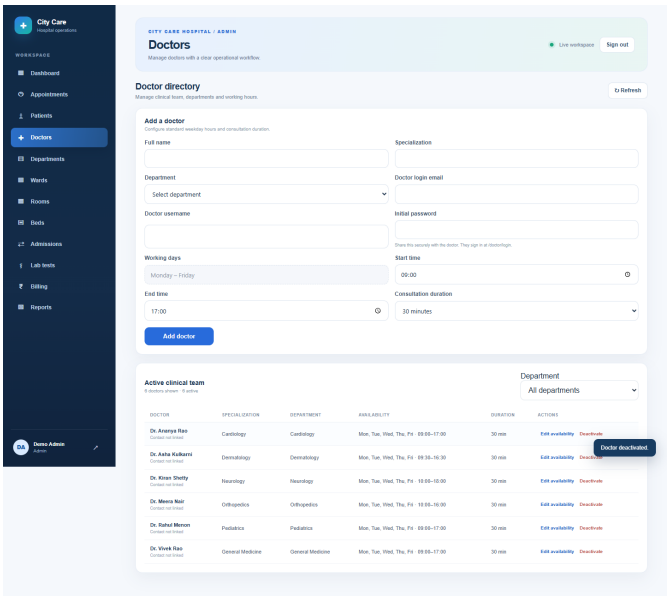


Figure 4. Doctor creation, duration update, and deactivation workflow.

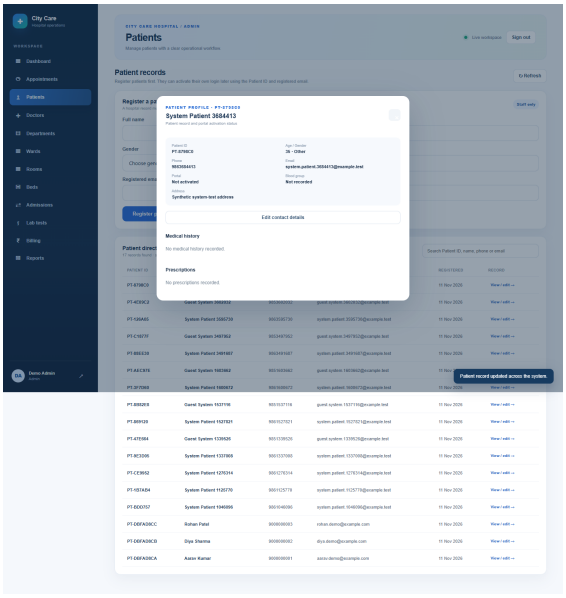


Figure 5. Patient registration and persisted update.

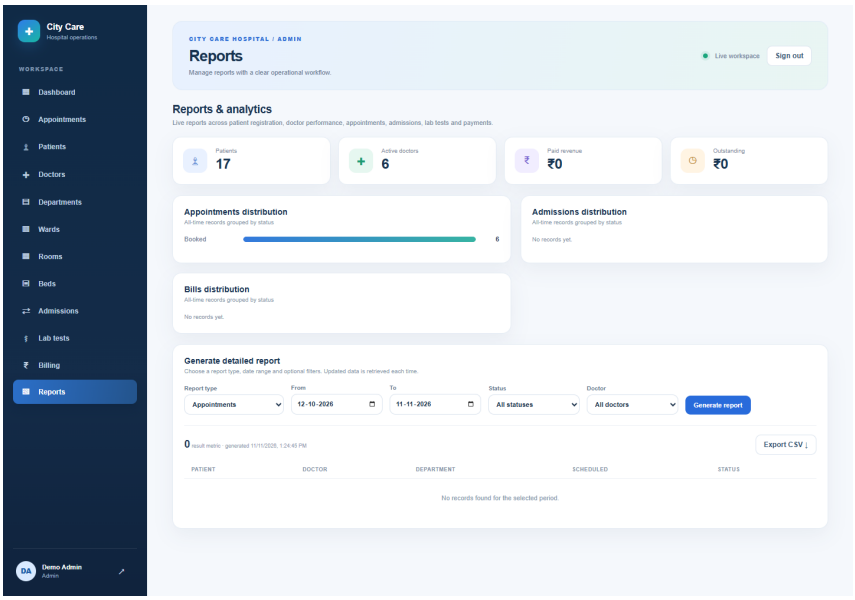


Figure 6. Admin reporting workspace with persisted data.

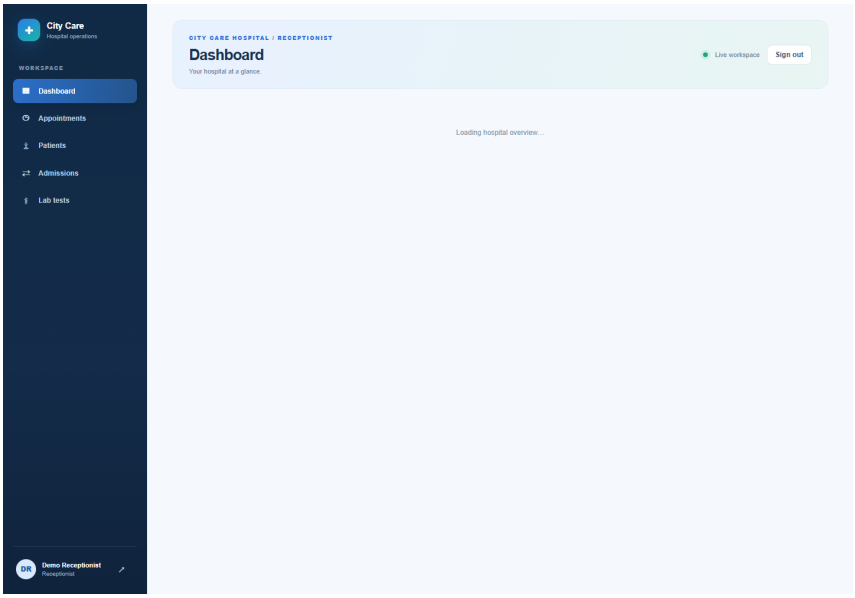


Figure 7. Receptionist dashboard and role-restricted navigation.

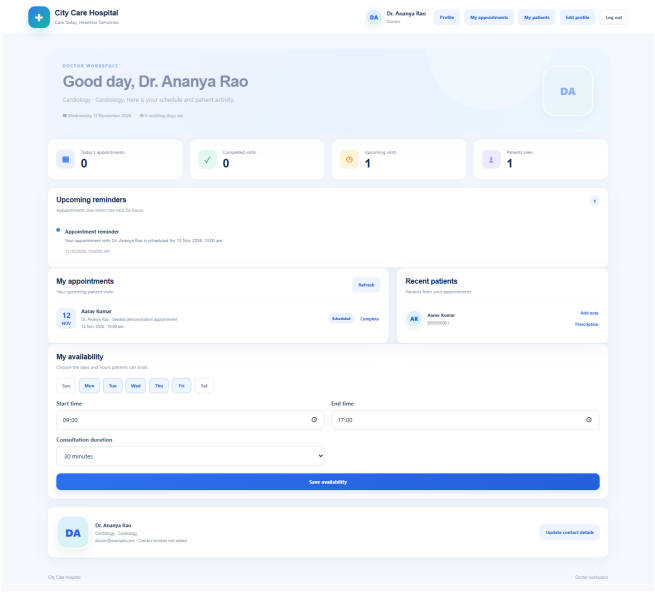


Figure 8. Doctor dashboard rendered after PageHeader fix.

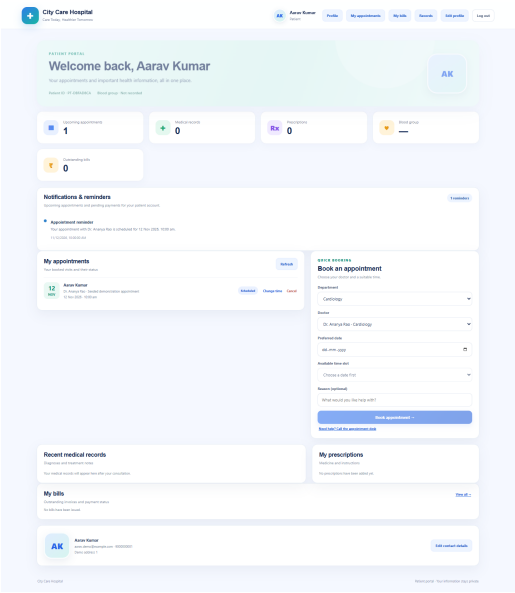


Figure 9. Patient dashboard and scoped portal rendered.

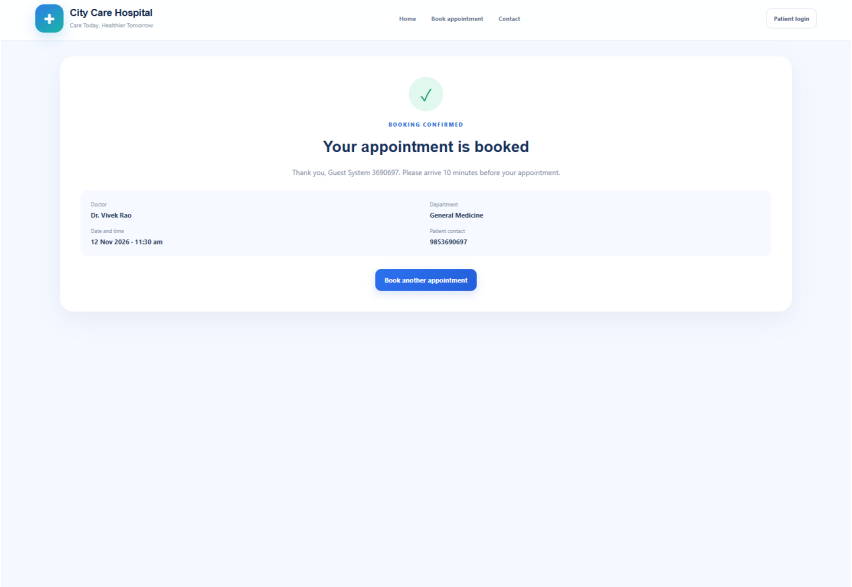


Figure 10. Guest booking confirmation.

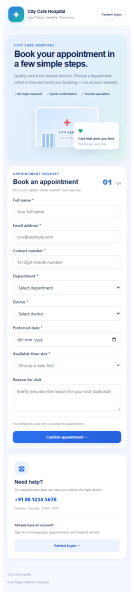


Figure 11. Guest booking page at 390px mobile viewport.

14. Requirement-to-Test Traceability

Requirement area	Mapped tests	Coverage outcome
FR-01 Identity and RBAC	UT identity/RBAC; IT-01; ST-AUTH-NEG-01, ST-ADMIN-01, ST-RECEPTION-01	Covered; PASS
FR-02 Patients	UT patient validation; IT-01/IT-04; ST-ADMIN-PATIENT-01, ST-PATIENT-01	Core covered; mutation gaps noted
FR-03 Doctors/departments	UT doctor/department validation; IT-01; ST-ADMIN-DEPT-01, ST-ADMIN-DOCTOR-01, ST-DOCTOR-01	Core covered; doctor self-update gap
FR-04 Appointments	UT slot/availability; IT-02; ST-GUEST-01, ST-GUEST-NEG-01	API/guest covered; role UI mutations pending
FR-05 Admissions/facilities	UT admission/bed/discharge; IT-03	API covered; full UI pending
FR-06 Clinical/laboratory	UT clinical rules; IT-04/IT-06; ST-DOCTOR-01, ST-PATIENT-01	Visibility covered; clinical UI mutation pending
FR-07 Billing/payments	UT calculations/payment; IT-05; ST-PATIENT-01	API covered and scope fixed; UI payment pending
FR-08 Notifications/reports	UT notification rules; IT-07/IT-08; ST-ADMIN-REPORTS-01	Covered; PASS
NFR Validation/security	Unit + integration negative paths; role-denial system checks	Covered within tested scope
NFR Responsive/performance	ST-RESPONSIVE-01; build timing	Mobile smoke covered; load benchmark pending

Traceability is based on the implemented feature set and the repository's SRS/test-plan material. The Excel workbook provides the detailed test rows, preconditions, steps, expected and actual results, status, and evidence reference.

15. Limitations

- Twenty UI/manual cases listed in Section 9 were not executed; none is counted as passed.
- The browser suite used Chrome only. Cross-browser behavior, formal accessibility conformance, and assistive-technology operation were not certified.
- No production-scale load test, penetration test, vulnerability certification, disaster-recovery exercise, or sustained availability test was performed.
- The mobile check was a representative 390px viewport smoke test, not a full device laboratory.
- Browser print-dialog/receipt behavior was not automated because native print UI is outside the page automation surface.
- The tests used deterministic local data. Production integrations, email/SMS gateways, deployment infrastructure, backup/restore, and real operational volumes were outside scope.
- Expected console resource messages from deliberately rejected requests and restricted external resources were retained; the final run had no uncaught page error.

16. Conclusion

The Hospital Management System passed every test that was actually executed: 44 unit, 8 integration, and 12 system cases. Four defects were identified, corrected, and verified by regression. The isolated databases, raw logs, JSON results, screenshots, workbook, and this report provide an auditable evidence chain.

The complete project is not declared fully passed because 20 planned system/manual scenarios remain unexecuted. The evidence supports the conclusion: EXECUTED AUTOMATED SCOPE PASSED; COMPLETE ASSESSMENT PARTIALLY EXECUTED. The remaining UI workflows and non-functional certifications should be completed before a production-readiness claim.

Executed automated scope: PASS
Complete project assessment: PARTIAL - additional manual and extended testing required